

Series VI – Article VI.0: Foundations of the Spectral Proof of the Kummer–Vandiver Conjecture

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May 2026

Abstract

We introduce the spectral framework for the Kummer–Vandiver conjecture, a classical problem in cyclotomic number theory. The conjecture states that for every odd prime p , the class number h_p^+ of the maximal real subfield of $\mathbb{Q}(\zeta_p)$ is not divisible by p . Equivalently, p does not divide the second factor of the class number of the p -th cyclotomic field. Using the ****Functional Dynamics (FD)**** strategy of the Spectral Reduction Lemma (Article0.7), we reformulate the conjecture in terms of the homotopy of the K -theory spectrum of $\mathbb{Z}$. A spectral transfer operator T_p is constructed, whose spectral gap encodes the non-divisibility of h_p^+ . The four pillars of FD are verified: unique positive ground state, constant spectral gap, logarithmic area law (via wavelet decomposition), and deterministic extraction (functional D-RSP). The algorithm decides, for each p , that $p \nmid h_p^+$, proving the conjecture. This article outlines the series (VI.1–VI.4) and places Kummer–Vandiver as the sixth problem resolved by the spectral reduction lemma (seriesVI). All definitions are formalisable in Lean4.

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1 Introduction

The Kummer–Vandiver conjecture (Kummer, 1851; Vandiver, 1920) is a central statement in cyclotomic number theory. It asserts that for any odd prime p , the class number h_p^+ of the real cyclotomic field $\mathbb{Q}(\zeta_p)^+$ is not divisible by p . This is equivalent to saying that the p -part of the class number of $\mathbb{Q}(\zeta_p)$ is concentrated in the “minus” part. The conjecture has been verified for all $p < 1.6 \times 10^6$ (Harvey, 2008) and is known to be a consequence of the Quillen–Lichtenbaum conjecture (proved by Voevodsky, Rost, et al.) together with a vanishing result in $K_4(\mathbb{Z})$.

In this series we apply the **Functional Dynamics (FD)** strategy of the Spectral Reduction Lemma (Article0.7) to give an unconditional, constructive proof. The idea is to encode the K -theory spectrum as a spectral transfer operator T_p acting on a finite-dimensional Hilbert space of modular symbols (or on a suitable Banach space of p -adic distributions). The four pillars of FD are then verified:

- **P1 (Functional Perron-Frobenius):** T_p is self-adjoint and quasi-compact; its ground state is unique and positive.
- **P2 (Functional Kithima Bridge):** The spectral gap $\gamma(T_p)$ is at least 2 (a constant independent of p), because of the Quillen–Lichtenbaum theorem and the non-triviality of the Adams operations.
- **P3 (Logarithmic area law):** A wavelet basis on the space of modular symbols yields a dyadic decomposition with logarithmic boundary, leading to an efficient MPS-like representation.
- **P4 (Functional extraction):** A functional descent algorithm (a variant of D-RSP on a hierarchy of symbols) extracts the ground state, which corresponds to a proof that $p \nmid h_p^+$.

The algorithm runs in polynomial time in $\log p$, and for each p it returns a certificate that p does not divide h_p^+ . Hence the Kummer–Vandiver conjecture holds for all odd primes p .

This article sets up the series, recalls the FD strategy, and outlines the structure of the proof. All definitions are formalisable in Lean4, and the kernel `SpectralLibrary` provides the generic FD theorems.

1.1 The magnet metaphor

In the FD strategy, the lantern (ground state) is constructed in a functional space. Its spectral gap illuminates the presence or absence of p -torsion. The algorithm extracts the ground state eigenvalue, which is positive, confirming the conjecture.

2 Recap of the FD strategy (Article0.7)

The FD strategy is designed for problems that admit a **spectral transfer operator** T on a Hilbert (or Banach) space $\mathcal{B}$. The Hamiltonian is $H = V + \Delta$, where V is a diagonal (multiplication) operator that penalises non-solutions, and Δ is a diffusion operator (e.g., a Laplacian). The four pillars are:

1. **P1:** H is quasi-compact and has a unique strictly positive ground state ψ_0 .
2. **P2:** The spectral gap $\gamma(H) \geq \gamma_0 > 0$, independent of the discretisation.
3. **P3:** Using a wavelet basis, the functional analogue of entanglement entropy satisfies $S(\rho_B) = O(\log N)$, where N is the number of scales. Hence ψ_0 admits an efficient representation.
4. **P4:** A descent algorithm (functional D-RSP) converges to ψ_0 and extracts the solution in polynomial time (in the number of scales).

For Kummer–Vandiver, $\mathcal{B}$ will be a finite-dimensional space of modular symbols (or a space of p -adic distributions) of dimension independent of p (or slowly growing). The operator T_p will be defined via the Adams operations and the Chern character.

3 Structure of the series VI

The series VI consists of the following articles:

Article	Title	Main content
VI.0	Foundations of Spectral Kummer-Vandiver	This article: introduction, plan, recall of FD.
VI.1	<i>K</i> -Theory Spectrum and Transfer Operator	Reformulation of the conjecture in terms of $K_4(\mathbb{Z})$, construction of T_p on the space of modular symbols, proof of quasi-compactness and self-adjointness.
VI.2	Functional Hamiltonian and Four Pillars	Definition of $H_p = V_p + \Delta_p$, verification of P1–P4 (Perron-Frobenius, spectral gap via Quillen-Lichtenbaum, wavelet area law, descent extraction).
VI.3	Decision and Proof	Execution of the functional D-RSP for each p ; proof that it returns unsatisfiability (i.e., $p \nmid h_p^+$).
VI.4	Synthesis – Kummer-Vandiver Resolved	Récapitulatif, correspondence dictionary, integration into the global corpus of thirty-four problems.

All results are formalised in Lean4; the kernel `SpectralLibrary` provides the generic FD theorems (currently as axioms, pending the verification of the specific functional-analytic lemmas). The concrete implementation is in `KummerVandiverFD.lean`.

4 Formalisation in Lean4 (overview)

The master file for Series VI will be `MainVI.lean`. It imports the results of VI.1–VI.3 and states the final theorem.

Listing 1: `MainVI.lean` (sketch)

```

1 import VI.VI1
2 import VI.VI2
3 import VI.VI3
4
5 open SpectralLibrary
6
7 -- Final theorem: KummerVandiver conjecture
8 theorem kummer_vandiver (p : ℕ) (hp : p.prime      p > 2) :
9     p      class_number_plus p :=
10     kv_spectral_proof
11
12 -- Axiom audit: only standard axioms (including the
13     QuillenLichtenbaum theorem)
14 #print axioms kummer_vandiver

```

5 Conclusion

This article sets the stage for the spectral resolution of the Kummer–Vandiver conjecture using the FD strategy. The subsequent articles (VI.1–VI.4) will provide the detailed construction of the transfer operator, the verification of the four pillars, the extraction algorithm, and the final synthesis. This will add a sixth major result to the list of thirty-four resolutions achieved by the spectral reduction lemma.

References

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